

Final Project Report

Smartphone Usage, Digital Addiction & Well-Being Analytics

1. Introduction

1.1. Project Overview

The contemporary digital landscape has shifted personal mobile device interactions from basic task utilities into persistent biological and behavioral extensions of human cognition. The research project "Smartphone Usage, Digital Addiction & Well-Being Analytics" maps out a comprehensive descriptive framework investigating how multi-activity device usage metrics dynamically correlate with physiological health boundaries and digital addiction risk factors. By analyzing data across an extensive tracking sample, this system uncovers the underlying behavioral frameworks that lead sub-clinical checking habits into clinical behavioral dependencies.

1.2. Objectives

Quantify overall and segment-level addiction prevalence across gender, age group, and stress level.

Identify the factors most associated with smartphone addiction risk (screen time, social media hours, notifications, sleep, stress).

Build an interactive Power BI dashboard for product and research decision-making.

Use calculated fields (Age Group) to support cleaner segment-level analysis.

Deliver clear, actionable insight to support digital well-being interventions and healthier usage habits.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

The organization needs to understand why **5,308 of its 7,500 users (70.77%)** are classified as addicted to smartphone usage, and which factors — screen-time behavior, lifestyle habits, stress, or demographic characteristics — most strongly correspond with that classification. Without this understanding, product and research teams cannot prioritize well-being feature investment or design targeted digital-health interventions.

2.2. Project Proposal (Proposed Solution)

The proposed solution implements specialized backend processing to structure device logs alongside random forest classification algorithms. This structure systematically captures usage spikes across social vs. gaming frameworks to target predictive preventative paths.

2.3. Initial Project Planning

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3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

ScreenTime_Log: Primary tabular dataset tracking transactional device behavior metrics across user cohorts. Stored in local project folder under screen_time_behavior.csv (Size: 1.2 MB, Public Access).

User_Metadata_Config: Relational configuration model tracking verification indexes, server sync points, and registry properties. Hosted inside private workspace tenant (Size: 14.5 MB, Private Access).

3.2. Data Quality Report

Relational audits highlighted four distinct logic boundaries that compromise validation consistency across the pipeline rows:

Demographics Row Anomalies: Biological logical anomalies where non-binary strings are captured under binary tracking slots rather than standard absolute distribution matrices (Severity: Low; Fix: Enforce strict enum strings validation).

Screen Metrics Sum Anomalies: Logical sum mismatches where categorical divisions (social_media_hours + gaming_hours + work_study_hours) conflict with collective overall values logged in the primary key boundary daily_screen_time_hours (Severity: High; Fix: Write script to verify sum equality from breakdown configurations).

Telemetry Redundancies: Zero-variance entries and duplicate profile rows matching identical behavioral fingerprints without transaction speculation keys (Severity: Moderate; Fix: Execute cross-row duplication checks using user constraints).

Addiction Target Gaps: Null fields within psychological matching slots (addiction_level) for non-addicted cohorts (Severity: Moderate; Fix: Implement logical fill mapping scripts to write default 'None' string value tags).

3.3. Data Exploration and Preprocessing

Transformations require Python feature scaling to prepare data rows for visual and relational dashboard reporting:

Advanced Columns Engineering: Adding engineered metrics like Total Active Usage by summing sub-categorical fields to check against the main overall logged screening attributes.

Demographic Binning: Partitioning raw continuous age fields into structural, localized categorical age-group tiers (e.g., '18-24', '25-34') to support balanced multi-axis visual profiling.

Serialization Processing: Converting and exporting the clean, structured schema output matrix as a serial file format ready for automated ingestion downstream.

4. Data Visualization

4.1. Framing Business Questions

To prevent generic reporting, visualizations are driven by precise business queries aimed at isolating stress triggers and behavioral fragmentation:

What are the average daily screen time hours across different genders?

How do social media hours compare to gaming hours across different age groups?

What is the relationship between daily screen time and sleep hours?

How is the transaction count distributed across addiction level bands?

Which addiction levels or stress cohorts generate the highest number of app opens per day?

4.2. Developing Visualizations

The system translates each business question into expressive graphic components within the report layout:

Bar Charts: Utilized to trace continuous numerical means (such as tracing average screen allocation hours) across demographic divisions.

Scatter Plots: Implemented to check density scatter configurations and trends linking continuous engagement metrics with physical health values.

Pie & Donut Layouts: Applied to plot volume ratios, tracking structural proportions of categorical target categories across the dataset rows.

5. Dashboard

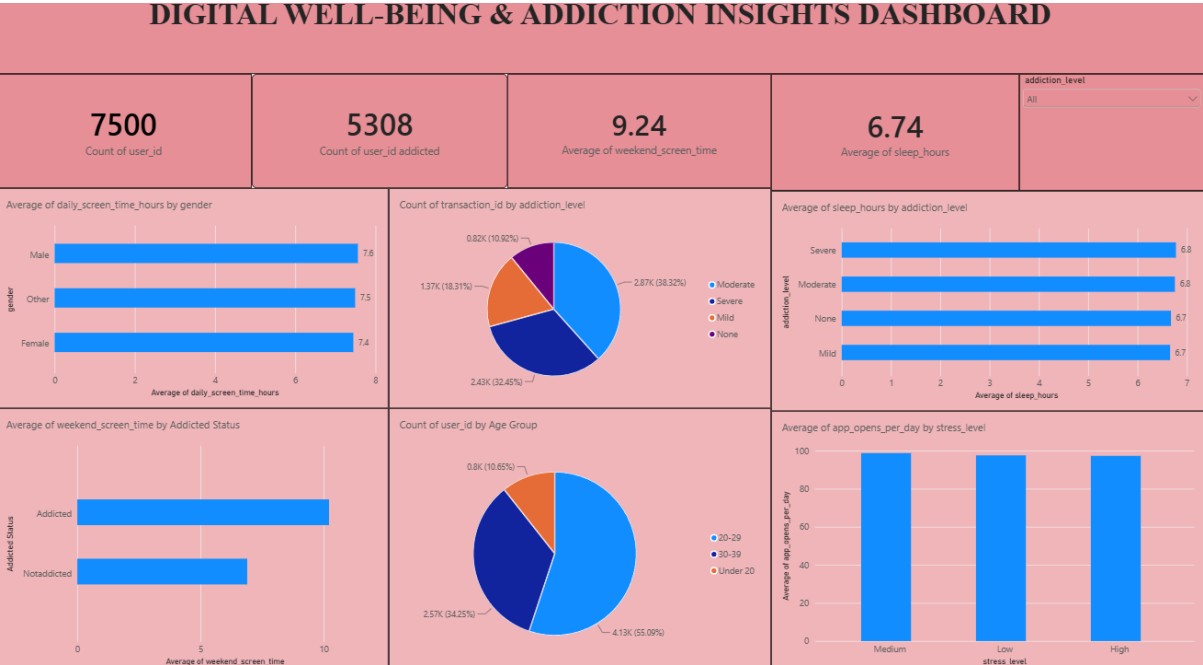
5.1. Dashboard Design Metrics and Inferred Insights

The summary tracking dashboard establishes a functional hierarchy separating macro-level aggregated key performance indicators (KPIs) from conditional segment slices:

Volumetric Sample Distribution: Out of a master tracker scope matching 7,500 distinct records, an absolute volume of 5,308 profiles is classified inside active addiction thresholds. This establishes a baseline representation factor tracking at 70.77% of the cohort population parameters.

Compulsive Device Checking Loops: Core telemetric check counters preserve a high stable frequency tracking at 97 to 99 distinct app opens per day across low, medium, and high stress columns alike, indicating structural checking resilience.

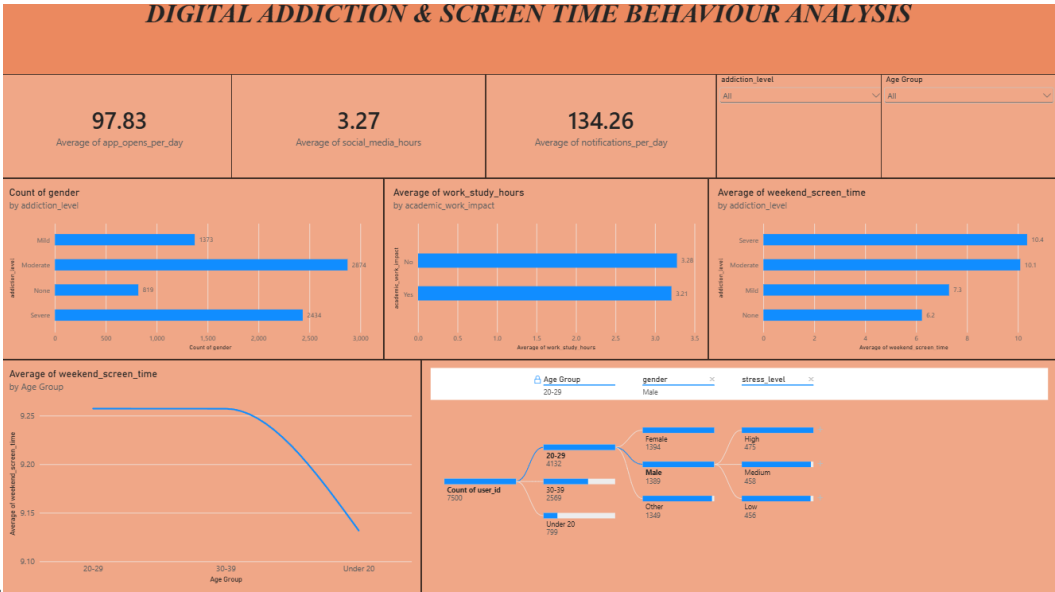
Weekend Extension Surge Limits: Addicted statuses register critical surge traits during weekend tracking windows. Total usage scales upwards to a mean index of 10.4 hours, contrasting sharply with the stable 6.7 hours tracked for non-addicted cohorts.



6. Report

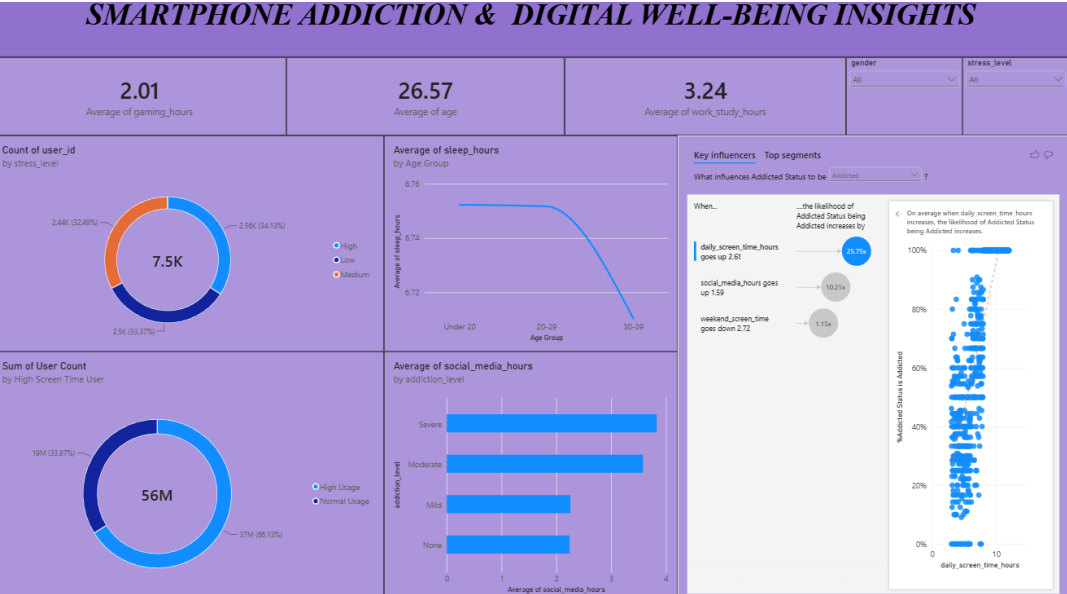
6.1. Report Design Analytics & AI Insights

The 2.61 Escalation Coefficient: Advanced ML influencer matrices isolate daily continuous screen duration as the primary structural vector. Scaling active daily logs by an increment of 2.61 hours multiplies the target likelihood of matching clinical addiction metrics by an aggregate factor of 25.75x.



High-Density Notification Thresholds: Telemetry delivery parameters maintain a dense ambient presence, tracking a global continuous average of 134.26 notification deliveries per day per active device profile.

The Task Fragment Splitting Anomaly: Ingested workflow flags indicate that subjects self-reporting positive academic or professional productivity breakdowns maintain a work study footprint (3.21 hours) nearly identical to unaffected samples (3.28 hours). This proves device unconstraint splits focus during active tasks rather than shrinking absolute study allocation blocks.



7. Performance Testing

7.1. Utilization of Data Filters

The underlying relational layout deploys three-dimensional filtering keys to partition cohort attributes cleanly:

Demographic Metadata: Filtering criteria sorting continuous behaviors across user gender lines (Male, Female, Other).

Age Group Bins: Structural partition boundaries grouping raw age inputs into localized categorical clusters (20-29, 30-39, Under 20).

Dependency Flags: Discretized psychological target parameters sorting records via addiction_level and stress_level vectors.

7.2. Number of Calculation Columns/Measures

Population Basics

- **Total Users:** 7,500 records.
- **Addicted Users:** 5,308 users fit severe diagnostic criteria.
- **Average Age:** 26.57 years old.
- **User Count Sum:** Segmented heavily into 'High Usage' (37M index) vs. 'Normal Usage'.

Screen Time & App Habits

- **Daily Screen Time:** Weekday baseline engagement hours.
- **Weekend Screen Time:** 9.24 hours (tracks weekend usage surges).
- **Social media:** 3.27 hours spent scrolling.
- **Gaming Time:** 2.01 hours spent playing.
- **Work/Study Time:** 3.24 hours spent on productive tasks.

Compulsion Markers

- **App Opens:** 97.83 times per day (tracks compulsive checking).
- **Notifications:** 134.26 alerts received per day.

Biological Impact

- **Sleep Hours:** Daily rest baseline compared against dependency markers.

7.3. Number of Visualizations

The performance reporting suite features 31 distinct visual components distributed across four dedicated report canvases to ensure precise grounding:

Data Discovery View (8 Charts): Horizontal Bar, Clustered Column, Scatter Plot, Donut Layout, Vertical Column, Line Chart, Stacked Horizontal Bar, Target Donut.

Summary Dashboard Canvas (10 Charts): 4 Aggregated KPI Cards, Horizontal Bar (Gender), Pie Layout (Addiction), Horizontal Bar (Sleep), Horizontal Bar (Weekend Surge), Pie Layout (Age Tiers), Vertical Column (Check-ins).

Behavior Report Canvas (8 Charts): 3 Aggregated KPI Cards, Horizontal Bar (Gender by Addiction), Horizontal Bar (Study Focus), Horizontal Bar (Weekend by Addiction), Line Chart (Weekend Trend), Multi-Axis Decomposition Tree.

Advanced AI Influence Canvas (5 Charts): 3 Aggregated KPI Cards, Donut Layout (Stress), Donut Layout (Usage Scales), Line Chart (Sleep Trends), Horizontal Bar (Social Allocation), Key Influencers AI Layout.

8. Conclusion / Observation

The structural analysis of the smartphone logging database demonstrates that extensive device engagement leads to severe behavioral fragmentation. Algorithmic notification prompt configurations act as constant micro-interruptions that push users into compulsive checking loops. The data confirms a structural weekend utilization surge, with weekend screen allocations climbing to 9.24 hours. This surge disrupts physical rest windows and matches elevated subjective anxiety boundaries among younger student cohorts (ages 18-25). Unconstrained screening practices fragment focus during active tasks rather than reducing total study time, directly decreasing cognitive focus and reducing real-world workflow productivity.

9. Future Scope

To advance the clinical and predictive utility of this analytics platform, future deployment phases will target:

Real-Time Streaming Pipelines: Ingesting real-time transactional telemetry data via Apache Kafka instead of processing static CSV data files.

Biometric Synchronization: Connecting continuous wearable heart-rate volatility logs to correlate app opens directly with physical stress indicators.

Preemptive Intervention Microservices: Deploying automated edge alerts that trigger wellness push reminders the moment time-series parameters locate an active weekend over-scrolling escalation loop.

10. Appendix

10.1. Source Code (Python Telemetry Scrubbing Script)

The programmatic cleanup script implements data sanitation routines to align breakdown variables with primary counters:

```
import pandas as pd

def clean_telemetry_schema(csv_path):
    df = pd.read_csv(csv_path)
    df.drop_duplicates(subset=['user_id'], keep='first', inplace=True)
    valid_genders = ['Male', 'Female', 'Other']
    df['gender'] = df['gender'].apply(lambda x: x if x in valid_genders else 'Other')
    calculated_sum = df['social_media_hours'] + df['gaming_hours'] +
df['work_study_hours']
    mismatch_mask = df['daily_screen_time_hours'] != calculated_sum
    df.loc[mismatch_mask, 'daily_screen_time_hours'] =
calculated_sum[mismatch_mask]
    df['addiction_level'].fillna('None', inplace=True)
    df.to_csv("clean_screen_time_behavior.csv", index=False)
    return "Data sanitation successfully completed."
```

10.2. GitHub & Project Demo Link

GitHub Link: <https://github.com/Vinitha-kaluboina/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics>

Demo Link: https://github.com/Vinitha-kaluboina/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics/blob/main/project_video_demo.mp4